

API Testing Guide Using Postman & RestAssured

Here are some commonly used HTTP status codes and their meanings, along with examples of valid live APIs for testing:

Common HTTP Status Codes and Their Meanings

1. 200 OK
 - Meaning: The request was successful, and the server returned the expected response.
 - Example: A successful GET request to retrieve data.
 - Live API Example :
 - `GET https://jsonplaceholder.typicode.com/posts/1`
 - This returns a specific post from a sample JSON API.
2. 201 Created
 - Meaning : The request has been fulfilled, and a new resource was created.
 - Live API : A POST request that creates a new resource on the server.
 - :
 - `POST https://jsonplaceholder.typicode.com/posts`
 - Sending JSON data in the body to create a new post.
3. 204 No Content
 - Meaning : The request was successful, but there is no content to return.
 - Live API : A DELETE request where the resource was deleted successfully.
 - :
 - `DELETE https://jsonplaceholder.typicode.com/posts/1`
 - This deletes a specific post and returns no content.
4. 400 Bad Request
 - Meaning : The server could not understand the request due to invalid syntax.
 - Live API : A malformed request that the server can't process.
 - :
 - `GET https://jsonplaceholder.typicode.com/posts/abc`
 - Trying to request a post with an invalid ID will return a 400 error.
5. 401 Unauthorised
 - Meaning : Authentication is required, and the user is not authenticated.
 - Live API : Accessing a protected resource without valid credentials.
 - :
 - `GET https://reqres.in/api/secure-endpoint`
 - Accessing a secure endpoint without proper authentication.
6. 403 Forbidden
 - Meaning : The request is valid, but the server refuses to authorize it.
 - Live API : Trying to access a restricted resource.
 - :

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- GET <https://jsonplaceholder.typicode.com/admin>

- Accessing a restricted resource that you don't have permission to access.

7. 404 Not Found

- Meaning : The server cannot find the requested resource.
- Live API : A GET request to a non-existent endpoint or resource.
- :

- GET <https://jsonplaceholder.typicode.com/posts/12345>

- Requesting a post that does not exist.

8. 500 Internal Server Error

- Meaning: The server encountered an unexpected condition that prevented it from fulfilling the request.
- Example: A server misconfiguration or issue causes the error.
- Live API Example :

- GET <https://httpstat.us/500>

- This returns a 500 error to simulate a server issue.

9. 502 Bad Gateway

- Meaning: The server, while acting as a gateway or proxy, received an invalid response from the upstream server.
- Example: A server in the middle cannot process the request properly.
- Live API Example :

- GET <https://httpstat.us/502>

- This returns a 502 error to simulate a bad gateway issue.

10. 503 Service Unavailable

- Meaning: The server is temporarily unable to handle the request due to maintenance or overload.
- Example: A server under maintenance or experiencing high load.
- Live API Example :

- GET <https://httpstat.us/503>

- This returns a 503 error to simulate a service unavailability.

Live APIs for Testing:

1. JSONPlaceholder API :

- A free online REST API for testing and prototyping.
- Base URL : <https://jsonplaceholder.typicode.com/>
- Example Endpoints:

- GET <https://jsonplaceholder.typicode.com/posts>

- POST <https://jsonplaceholder.typicode.com/posts>

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2. ReqRes:

- A hosted REST API that simulates user management scenarios.
- Base URL : <https://reqres.in/>
- Example Endpoints:
 - GET <https://reqres.in/api/users>
 - POST <https://reqres.in/api/users>

3. HTTP Stat.us :

- A simple service to return different HTTP status codes for testing.
- Base URL : <https://httpstat.us/>
- Example Endpoints:
 - GET <https://httpstat.us/200>
 - GET <https://httpstat.us/404>

4. The Dog API :

- A fun API that returns pictures and information about dogs.
- Base URL : <https://thedogapi.com/>
- Example Endpoints:
 - GET <https://api.thedogapi.com/v1/breeds>
 - GET <https://api.thedogapi.com/v1/images/search>

6 step-by-step practical guide to API testing, with examples using Postman and RestAssured.

1. Understanding API Basics

API stands for Application Programming Interface, a set of rules that define how software components should interact. APIs typically use HTTP requests to communicate, which can involve the following methods:

- GET: Retrieve data from a server.
 - POST: Send data to a server to create a new resource.
 - PUT: Update an existing resource.
 - DELETE: Remove a resource from the server.
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2. Setting Up Tools for API Testing

For this guide, we'll use two common tools:

- Postman: A user-friendly GUI tool for testing APIs.
- Rest Assured: A Java-based API testing library for automation testing.

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3. Using Postman for API Testing

Step 1: Install Postman

Download and install Postman from Postman's official website.

Step 2: Create a New Request

- Open Postman, and click New > Request.
- Select the request type (e.g., GET, POST) and enter the API URL. For example:
 - GET `https://jsonplaceholder.typicode.com/posts/1`

Step 3: Add Parameters/Body

- If needed, add query parameters or request body in Params or Body tabs.

Step 4: Send Request

- Click Send and examine the response (e.g., Status Code, JSON data).

Step 5: Writing Tests

- In the Tests tab, write simple assertions using JavaScript:

javascript

```
pm.test("Status code is 200", function () {  
  
    pm.response.to.have.status(200);  
  
});
```

Step 6: Collection Runs and Automation

- Group your requests into collections and automate them using Postman's Collection Runner for repeated testing.
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4. Using Rest Assured for API Automation

Rest Assured is a powerful tool for automating API tests in a Java environment.

Step 1: Set Up Rest Assured

Add Rest Assured as a dependency in your `pom.xml` (for Maven projects):

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xml

```
<dependency>

    <groupId>io.rest-assured</groupId>

    <artifactId>rest-assured</artifactId>

    <version>4.4.0</version>

    <scope>test</scope>

</dependency>
```

Step 2: Writing a Basic Test

Here's a simple example of testing a GET request using Rest Assured:

java

```
import io.restassured.RestAssured;

import io.restassured.response.Response;

import static io.restassured.RestAssured.*;

import static org.hamcrest.Matchers.*;

public class ApiTest {

    public static void main(String[] args) {

        RestAssured.baseURI =
"https://jsonplaceholder.typicode.com";

        // Simple GET request

        given()

            .when()

            .get("/posts/1")
```

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```
.then()  
  
.statusCode(200)  
  
.body("userId", equalTo(1));  
  
}  
  
}
```

Step 3: Running the Test

- Use a testing framework like JUnit or TestNG to run your test cases.
 - Execute the test with assertions to validate the status code and response body.
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5. Key API Testing Concepts

1. Status Code Validation: Verify the status code in each response. Common status codes include:
 - 200 OK: Request was successful.
 - 201 Created: Resource created successfully.
 - 400 Bad Request: Invalid request syntax.
 - 401 Unauthorised: Authentication required.
 - 500 Internal Server Error: Server encountered an error.
2. Response Body Validation: Ensure the returned data matches expectations. Use assertions to check specific fields:

java

```
body("title", equalTo("foo"))
```

3. Header Validation: Validate HTTP headers for security and content:

java

```
header("Content-Type", "application/json; charset=utf-8")
```

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4. Authentication : Some APIs require authentication. Add authorization tokens in the Headers section in Postman or pass tokens in Rest Assured:

java

```
given().auth().oauth2("YOUR_ACCESS_TOKEN")
```

5. Negative Testing : Send invalid data to test how the API handles errors.
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6. Advanced Techniques

1. Parameterized Tests: Reuse test cases with different sets of input data using variables or environments.
 2. Data-Driven Testing: Use CSV or JSON files to input multiple datasets in automated tests.
 3. Mocking APIs: Use tools like Postman Mock Servers or WireMock to simulate APIs that may not be fully developed yet.
-

7. Reporting and Continuous Integration

- Use tools like Newman (Postman's command-line runner) to integrate API tests into your CI/CD pipelines.
 - For Rest Assured, combine with reporting frameworks like Extent Reports for detailed results.
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8. API Testing Best Practices

1. Clear Test Cases: Ensure your test cases cover all scenarios, including happy paths and edge cases.
 2. Independent Tests: API tests should not depend on each other to avoid false negatives.
 3. Monitor APIs: After testing, set up API monitoring using tools like Postman Monitors to catch downtime or slow performance.
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Conclusion

API testing is critical to ensuring that your backend services perform reliably. Whether you use Postman for manual testing or automate your API tests with Rest Assured, following this guide will help ensure that your APIs meet functionality, performance, and reliability expectations.